JIALE JIANG

Mechanical Engineering Student | Robotics Systems Developer

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ENGINEERING PROFILE

Mechanical engineering student building a 12-joint quadruped robot from custom actuators through C++ control, with hands-on experience in CAD, fabrication, electronics, testing, and technical leadership.

EDUCATION

Mount San Antonio College | Walnut, CA **2023–Present**

Mechanical Engineering, transfer preparation | Cumulative GPA: 3.61 | Major GPA: 4.0|President's List: Fall 2025, Spring 2026

Coursework: Calculus I-III, Linear Algebra & Differential Equations, Physics I-II, Statics, Engineering Graphics. In progress: Dynamics, MATLAB, Materials.

TECHNICAL SKILLS

Design & Build: SolidWorks, ANSYS, CFD, 3D printing, machining, welding, laser cutting
Robotics & Code: C++, Python, MATLAB, embedded C, motor control, CAN, IMU sensors, Git/GitHub
Electronics & Test: STM32, KiCad, BLDC motors, encoders, PCB assembly and rework, soldering, bench testing

SELECTED ENGINEERING PROJECTS

12-DOF Quadruped Robot System | *Project Lead & Robotics Developer* **Sep 2025-Present**

- Designed and 3D-printed the body, legs, and linkages; used FEA to strengthen load-bearing parts before assembly.
- Built the leg-motion math and C++ software for homing, gait timing, motor faults, data logging, and IMU leveling over CAN.
- Integrated a 17.18 kg prototype and validated coordinated leg motion plus limited backward/crawl movement; a second design predicts 16.8% lower mass.
- Led teammates in firmware, soldering, and assembly, and kept setup and test notes in GitHub.

3D-Printed Cycloidal Robotic Actuator | *Mechanical Designer & Test Engineer* **Sep 2025-Present**

- Built more than 20 printable actuator revisions, using simulation and physical tests to improve shaft support, fit, and load paths.
- Built a dual-actuator test stand; one preliminary data set measured 14.7% median end-to-end efficiency. Integrated the motor, encoder, bearings, controller, and cooling into robot joints.

STM32 Motor Controller PCB & CAN Integration | *PCB & Embedded Systems Developer* **Sep 2025-Present**

- Co-designed the KiCad schematic and board layout, then assembled, soldered, reworked, and bench-tested the first board.
- Configured firmware, encoder feedback, CAN, limits, and safety features. The board did not become reliable or reach the robot, so I documented the faults and used commercial controllers.

Remote-Controlled Underwater ROV Thruster | *Mechanical Design Lead* **Jun 2025-Oct 2025**

- Designed a ducted propeller, compared versions with CFD, fabricated and tested prototypes, and achieved about 10 N of thrust; trained teammates in assembly and testing.

PROFESSIONAL EXPERIENCE

Living Spaces | *Sales Associate* | *Monrovia, CA* **May 2024-Nov 2024**

- Matched customer needs to practical options, explained tradeoffs, and coordinated complex orders across teams.

Floor & Decor | *Sales Associate* | *San Gabriel, CA* **Jan 2024-May 2024**

- Estimated material needs , explained installation requirements, and supported customers and daily inventory work.